

Muraleekrishna Gopinathan, PhD <https://linkedin.com/in/gmuraleekrishna>

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Perth, Western Australia

Senior-level AI/ML scientist with 5+ years of ML/data science expertise and 10+ years of software engineering knowledge. Proficient in productionisation of Generative AI solutions.

TECHNICAL PROFICIENCIES

Tools: AWS/Azure, Python, PyTorch, Bedrock, SageMaker, Foundry MATLAB, ROS, Git, Docker, Linux, ROS.

Skills: Generative AI, NLP, Programming, AI evaluation, Predictive Maintenance, Research, MLOps, Agentic AI, Computer Vision, Deep Learning, Robotics, System Engineering, Agile, Leadership, Communication.

PROFESSIONAL EXPERIENCE

Senior Data Engineer (Jun 2025 – Present) Fortescue Ltd, Australia

Agentic AI, LLM evaluation, AWS/Azure, OpenSearch, SolidJS, Zero, Tanstack, vLLM, AI architecture, Snowflake, LLM-as-a-judge.

- Conducted AI architecture design, code reviews, solution design decisions
- Developed Semantic search product RAG service processing 10M+ engineering documents
- Lead evaluation of contract-evaluation product improving AI feedback quality.

Lead Data Engineer (Apr 2020 – Jun 2025) Hitachi Rail STS, Australia

Fault identification, Data Science & Engineering, Machine learning, Data-lake, Analytics, Business Intelligence, Python, Apache Kafka, Starrocks, Postgres, MySQL, NumPy, Spark, MLFlow, Elasticsearch

- Improved quality of autonomous rail system maintenance by developing predictive models using Big Data platform.
- Developed machine learning models for predicting asset behaviour based on time-series data.
- Developed scalable and highly available ETL and Lakehouse architecture using Open-source ecosystem.
- Developed custom data parsers for decoding multi-format logs generated by autonomous locomotive computers. 80000+ lines of code.

Session Academic (Feb 2019 – Jun 2020) Australian National University, ACT

ROS, OpenCV, Verilog, FPGA, Control systems, Python, Systems Engineering

- Prepared course content and lab materials for students and recommend course improvements to academic committee.
- Tutored Systems Engineering and Embedded Systems courses.
- Successfully mentored 120+ students throughout completion of their academic project.

Software Engineer (Apr 2017– Oct 2017) Aconex (India) Pty Ltd, India

Ruby on Rails, Python, HTML, JavaScript, Agile Methodologies, Git, JIRA.

- Conceived and deployed multi-service large-scale application of over 1 million users.
- Designed new REST APIs, built UI components in Angular JS, improved hybrid mobile applications,
- CQRS-based micro-services using Ruby and JavaScript. 80000+ lines of code.
- Applied Test-Driven Development and Extreme programming methods for significantly improving software development process.

- Resolved a long existing bug and improved the performance and security of the application.

Software Programmer (Jun 2014 – Mar 2017)

Multunus Software Pty Ltd, Bangalore, India

Test Driven Development, Agile methodologies, Ruby, Postgres, HTML5, Git, JIRA.

- Led design, development, and testing of software products across multiple domains. Liaised with customers, users, as well as product owners, and identified business requirements.
 - Drove software development through agile methodology, and practiced pair programming as well as test driven development.
 - Built innovative intelligent recommendation system, increasing client's customer conversion ratio.
 - Delivered working software every 2 days, successfully managing very tight turnaround.
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RESEARCH WORK

StratXplore: Strategic Novelty-seeking and Instruction-aligned Exploration for Vision and Language Navigations: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024), Abu Dhabi, UAE, 2024. (Robotics, Computer Vision, Computational Language Modelling)*

Spatially-Aware Speaker for Vision Language Navigation Instruction Generation: *The 62nd Annual Meeting of the Association for Computational Linguistics (ACL), Bangkok, Thailand, 2024. (Generative AI, Transformers, Language Modelling, Computer Vision)*

Gesture Control of Boston Dynamic Spot using XRAI Vision Glasses for Defence Applications (2022): \$150,000 Grant project from Defence Science Centre, in collaboration with UWA, Chironix, Agili8 and Edith Cowan University. *(Robotics, Computer Vision)*

Indoor Semantic Scene understanding using 2d-3d Fusion: *2021 Digital Image Computing: Techniques and Applications (DICTA), Gold Coast, Australia, 2021. (Robotics and Computer Vision)*

Solving occlusion in object detection using multiple camera views (2019): *Developed deep learning model for object detection and tracking using multiple cameras. Improved detection accuracy and robustness. Master's Thesis. (Computer Vision)*

Altitude Control System Design for High Power Rockets (2019): *Designed optimal control scheme to achieve competition-winning apogee/altitude for high power rocket (HPR). (Systems modelling, Systems Engineering, Rocketry, Control systems)*

EDUCATION

Doctor of Philosophy, Computer Science

Scholarship

Edith Cowan University, Joondalup, Australia

Topics: Robotics, Computer Vision, Machine Learning, NLP, Generative AI

Jan 2021 – Sep 2025

Master of Engineering, Mechatronics Engineering

GPA 6.1/7

Australian National University, ACT, Australia

Topics: Robotics, Embedded systems, Computer Vision, Control systems

Feb 2018 - Dec 2019

Bachelor of Technology, Electronics and Communication Engineering

GPA 7.5/10

Mahatma Gandhi University, Kerala, India

Topics: Mechatronics, Control systems, Telecommunication systems

Jun 2010 - May 2014

ACHIEVEMENTS AND LEADERSHIP

- **Presented 2 International AI papers:** [IROS 2024](#) (Robotics, A rank) and [ACL 2024](#) (NLP, A* rank)
- **Team Lead:** Operation and Maintenance Data Science Team, Hitachi Rail STS, 2020.
- **Winner:** Vice Chancellors PhD by Research Scholarship, ECU, WA, 2021.
- **Team leader:** ANU Rocketry 2020 Avionics team, ANU, Canberra, 2019.